



9230 - The Aftermath of a Transient Deposition Event on Europa

Cycle: 3, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
January NIRSpec Observations of all 4 Phases of Europa				
	1	Europa leading NIRspec IFU high res	NIRSpec IFU Spectroscopy	(1) EUROPA
	2	Europa anti-Jovian NIRspec IFU high res	NIRSpec IFU Spectroscopy	(1) EUROPA
	3	Europa sub-Jovian NIRspec IFU high res	NIRSpec IFU Spectroscopy	(1) EUROPA
	4	Europa trailing NIRspec IFU high res	NIRSpec IFU Spectroscopy	(1) EUROPA
November NIRSpec Observations of all 4 Phases of Europa				
	5	Europa leading NIRspec IFU high res	NIRSpec IFU Spectroscopy	(1) EUROPA
	6	Europa anti-Jovian NIRspec IFU high res	NIRSpec IFU Spectroscopy	(1) EUROPA
	7	Europa sub-Jovian NIRspec IFU high res	NIRSpec IFU Spectroscopy	(1) EUROPA
	8	Europa trailing NIRspec IFU high res	NIRSpec IFU Spectroscopy	(1) EUROPA

ABSTRACT

Recent observations of Europa with JWST NIRSpec have revealed surprising surface compositional changes relative to previous infrared data and pinpoint the onset of changes to within the last several months. We request follow-up observations to monitor the aftermath of this clear compositional shift as the surface continues to evolve under the intense irradiation conditions at Europa's surface. Our proposed program will characterize the nature and extent of the modifications, while also providing a real-time laboratory on the ensuing radiolytic alteration chemistry happening on the surface. As changes are likely ongoing, it is imperative that such observations begin during the upcoming January Europa visibility window with JWST, necessitating Director's discretionary time.

OBSERVING DESCRIPTION

We request two sets of observations (4 rotational phases of Europa in each set) to obtain full NIRSpec G395H coverage of Europa's surface in each the next two observability windows (January and November 2025).

We replicate exactly the exposure parameters of the already successful observations from program 4023, which used a total integration time of 1288 seconds across 5 groups, 5 integrations, and 4 dithers with the NRSRAPID mode. As we are requesting that each set of 4 rotational Europa phases occur in a single JWST FOR, care must be taken to ensure that Jupiter does not fall on the MSA and cause leakage that will contaminate the spectra.

Doing a full visualization in Aladin using a fixed target at Europa's coordinates and position angle constraints from the moving target visibility tool reveals that there are multiple windows in January for which the trailing hemisphere of Europa can be observed without Jupiter falling on the MSA. The sub- and anti-Jovian geometries will be observed slightly toward Europa's leading side to ensure Jupiter is on the opposite side of Europa from the MSA.

The same Aladin analysis shows that there are two windows in November when Europa's leading side can be observed in JWST's leading FOR. Similarly, the sub- and anti-Jovian geometries will then be observed slightly toward Europa's trailing side to ensure Jupiter is on the opposite side of Europa from the MSA in November.

These two sets of observations will allow us to fully monitor Europa's surface composition as it continues to evolve following the apparent surface changes. If the January observations reveal all changes to already be reversed and the surface to have reverted to its pre-existing state, then we will relinquish any time given for November so that it may be used for other programs.

Proposal 9230 - Targets - The Aftermath of a Transient Deposition Event on Europa

Solar System Targets	#	Name	Level 1	Level 2	Level 3
	(1)	EUROPA	STD=JUPITER	STD=EUROPA	
	Comments: Extended=YES				

Proposal 9230 - Observation 1 - The Aftermath of a Transient Deposition Event on Europa

Observation	Proposal 9230, Observation 1: Europa leading NIRspec IFU high res											Fri Jan 03 21:00:10 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Europa leading NIRspec IFU high res (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2			Level 3		
	(1)	EUROPA	STD=JUPITER				STD=EUROPA					
	Comments: Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		4				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	5	5	false	true	NONE	4	20	1288.412	
Special Requirements	Between Dates 17-JAN-2025:00:00:00 and 17-FEB-2025:00:00:00 DEFAULT WINDOW: NOT OCCULTATION OF EUROPA BY JUPITER FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA FROM JWST LESS THAN 0.075 CENTRAL MERIDIAN LONGITUDE OF EUROPA FROM JWST BETWEEN 85 95 APPARENT DIAMETER EUROPA GREATER THAN 0.9											

Proposal 9230 - Observation 2 - The Aftermath of a Transient Deposition Event on Europa

Observation	Proposal 9230, Observation 2: Europa anti-Jovian NIRspec IFU high res Fri Jan 03 21:00:10 GMT 2025											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Europa anti-Jovian NIRspec IFU high res (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2			Level 3		
	(1)	EUROPA	STD=JUPITER				STD=EUROPA					
	Comments: Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		4				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	5	5	false	true	NONE	4	20	1288.412	
Special Requirements	Between Dates 17-JAN-2025:00:00:00 and 17-FEB-2025:00:00:00											
	DEFAULT WINDOW: NOT OCCULTATION OF EUROPA BY JUPITER FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA FROM JWST LESS THAN 0.075 CENTRAL MERIDIAN LONGITUDE OF EUROPA FROM JWST BETWEEN 146 166 APPARENT DIAMETER EUROPA GREATER THAN 0.9 SEPARATION OF EUROPA JUPITER FROM JWST GREATER THAN 27"											

Proposal 9230 - Observation 3 - The Aftermath of a Transient Deposition Event on Europa

Observation	Proposal 9230, Observation 3: Europa sub-Jovian NIRspec IFU high res											Fri Jan 03 21:00:10 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Europa sub-Jovian NIRspec IFU high res (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2			Level 3		
	(1)	EUROPA	STD=JUPITER				STD=EUROPA					
	Comments: Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		4				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	5	5	false	true	NONE	4	20	1288.412	
Special Requirements	Between Dates 17-JAN-2025:00:00:00 and 17-FEB-2025:00:00:00 SEPARATION OF EUROPA JUPITER FROM JWST GREATER THAN 27" CENTRAL MERIDIAN LONGITUDE OF EUROPA FROM JWST BETWEEN 24 44 APPARENT DIAMETER EUROPA GREATER THAN 0.9 DEFAULT WINDOW: NOT OCCULTATION OF EUROPA BY JUPITER FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA FROM JWST LESS THAN 0.075 NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA BY JUPITER FROM JWST											

Proposal 9230 - Observation 4 - The Aftermath of a Transient Deposition Event on Europa

Observation	Proposal 9230, Observation 4: Europa trailing NIRspec IFU high res											Fri Jan 03 21:00:10 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Europa trailing NIRspec IFU high res (Obs 4)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2			Level 3		
	(1)	EUROPA	STD=JUPITER				STD=EUROPA					
	Comments: Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		4				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	5	5	false	true	NONE	4	20	1288.412	
Special Requirements	Between Dates 17-JAN-2025:00:00:00 and 31-JAN-2025:00:00:00											
	SEPARATION OF EUROPA JUPITER FROM JWST GREATER THAN 27" CENTRAL MERIDIAN LONGITUDE OF EUROPA FROM JWST BETWEEN 265 275 APPARENT DIAMETER EUROPA GREATER THAN 0.9 DEFAULT WINDOW: NOT OCCULTATION OF EUROPA BY JUPITER FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA FROM JWST LESS THAN 0.075											

Proposal 9230 - Observation 5 - The Aftermath of a Transient Deposition Event on Europa

Observation	Proposal 9230, Observation 5: Europa leading NIRspec IFU high res											Fri Jan 03 21:00:10 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Europa leading NIRspec IFU high res (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2			Level 3		
	(1)	EUROPA	STD=JUPITER				STD=EUROPA					
	Comments: Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		4				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	5	5	false	true	NONE	4	20	1288.412	
Special Requirements	Between Dates 25-NOV-2025:00:00:00 and 30-NOV-2025:00:00:00											
	DEFAULT WINDOW: NOT OCCULTATION OF EUROPA BY JUPITER FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA FROM JWST LESS THAN 0.075 CENTRAL MERIDIAN LONGITUDE OF EUROPA FROM JWST BETWEEN 85 95 APPARENT DIAMETER EUROPA GREATER THAN 0.9 SEPARATION OF EUROPA JUPITER FROM JWST GREATER THAN 27"											

Proposal 9230 - Observation 6 - The Aftermath of a Transient Deposition Event on Europa

Observation	Proposal 9230, Observation 6: Europa anti-Jovian NIRspec IFU high res											Fri Jan 03 21:00:10 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Europa anti-Jovian NIRspec IFU high res (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2			Level 3		
	(1)	EUROPA	STD=JUPITER				STD=EUROPA					
	Comments: Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		4				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	5	5	false	true	NONE	4	20	1288.412	
Special Requirements	Between Dates 11-OCT-2025:00:00:00 and 01-DEC-2025:00:00:00 DEFAULT WINDOW: NOT OCCULTATION OF EUROPA BY JUPITER FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA FROM JWST LESS THAN 0.075 CENTRAL MERIDIAN LONGITUDE OF EUROPA FROM JWST BETWEEN 203 223 APPARENT DIAMETER EUROPA GREATER THAN 0.9 SEPARATION OF EUROPA JUPITER FROM JWST GREATER THAN 27"											

Proposal 9230 - Observation 7 - The Aftermath of a Transient Deposition Event on Europa

Observation	Proposal 9230, Observation 7: Europa sub-Jovian NIRspec IFU high res											Fri Jan 03 21:00:10 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Europa sub-Jovian NIRspec IFU high res (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2			Level 3		
	(1)	EUROPA	STD=JUPITER				STD=EUROPA					
	Comments: Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		4				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	5	5	false	true	NONE	4	20	1288.412	
Special Requirements	Between Dates 11-OCT-2025:00:00:00 and 01-DEC-2025:00:00:00 SEPARATION OF EUROPA JUPITER FROM JWST GREATER THAN 27" CENTRAL MERIDIAN LONGITUDE OF EUROPA FROM JWST BETWEEN 322 342 APPARENT DIAMETER EUROPA GREATER THAN 0.9 DEFAULT WINDOW: NOT OCCULTATION OF EUROPA BY JUPITER FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA FROM JWST LESS THAN 0.075 NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA BY JUPITER FROM JWST											

Proposal 9230 - Observation 8 - The Aftermath of a Transient Deposition Event on Europa

Observation	Proposal 9230, Observation 8: Europa trailing NIRspec IFU high res												Fri Jan 03 21:00:10 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Europa trailing NIRspec IFU high res (Obs 8)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(1)	EUROPA	STD=JUPITER				STD=EUROPA						
	Comments: Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		SMALL		1		4					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G395H/F290LP	NRSRAPID	5	5	false	true	NONE	4	20	1288.412		
Special Requirements	Between Dates 11-OCT-2025:00:00:00 and 01-DEC-2025:00:00:00												
	SEPARATION OF EUROPA JUPITER FROM JWST GREATER THAN 27" CENTRAL MERIDIAN LONGITUDE OF EUROPA FROM JWST BETWEEN 265 275 APPARENT DIAMETER EUROPA GREATER THAN 0.9 DEFAULT WINDOW: NOT OCCULTATION OF EUROPA BY JUPITER FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA FROM JWST LESS THAN 0.075												